

SECTION 1—APPLICABILITY

1.0 Definition of Dangerous Goods

Dangerous goods are articles or substances which are capable of posing a hazard to health, safety, property or the environment and which are shown in the list of dangerous goods in these Regulations or which are classified according to these Regulations.

Note:

Terms, which have special meaning within these Regulations, are defined in Appendix A.

1.1 Basis of these Regulations

- △ 1.1.1 The UN Subcommittee of Experts on the Transport of Dangerous Goods (SCoETDG) develops recommended procedures for the transport of all types of dangerous goods except radioactive materials. These procedures, applicable to all modes of transport, are published in the *Recommendations on the Transport of Dangerous Goods—Model Regulations*.

Note:

Recommendations on Tests and Criteria, which are incorporated into certain provisions of these Regulations are published as a separate manual (Recommendations on the Transport of Dangerous Goods, Manual of Tests and Criteria) available from the United Nations. This Manual includes:

- *Part I: Classification procedures, test methods and criteria relating to explosives of Class 1.*
- *Part II: Classification procedures, test methods and criteria relating to self-reactive and polymerizing substances of Division 4.1 and organic peroxides of Division 5.2.*
- *Part III: Classification procedures, test methods and criteria relating to articles or substances of Class 2, Class 3, Class 4, Division 5.1, Class 8 and Class 9.*
- *Part IV: Test methods concerning transport equipment.*
- *Part V: Classification procedures, test methods and criteria relating to sectors other than transport.*
- *Appendices: Information common to a number of different types of tests and national contacts for test details.*

- △ 1.1.2 The International Atomic Energy Agency (IAEA) develops recommended procedures for the safe transport of radioactive materials. These procedures are published in the *IAEA Regulations for the Safe Transport of Radioactive Material (2018 Edition)*. The requirements of these regulations as they pertain to air transport are reflected in Section 10.

1.1.3 The International Civil Aviation Organization (ICAO) has used these recommendations as the basis for developing the regulations for the safe transport of

dangerous goods by air by any aircraft (including both internal and external carriage). The ICAO regulations are codified in Annex 18 to the *Convention on International Civil Aviation* and in its *Technical Instructions for the Safe Transport of Dangerous Goods by Air* (Doc 9284 as amended) (Technical Instructions).

Note:

The term “aircraft” includes both aeroplanes and helicopters.

1.1.4 The IATA *Dangerous Goods Regulations* (the Regulations) contain all of the requirements of the Technical Instructions. IATA has included additional requirements, which are more restrictive than the Technical Instructions and reflect industry standard practices or operational considerations. These are identified by the symbol “” in the margin.

1.2 Application of these Regulations

STATE VARIATIONS: BHG-01, CAG-06, JMG-04, USG-01, VCG-02, VEG-04/08, ZAG-02

Editorial Note:

State and operator variations applicable to a subsection are indicated by the appropriate alpha-numeric code, as described in 2.8.2 and 2.8.4 respectively.

1.2.1 Applicability

The IATA Dangerous Goods Regulations are applicable to:

- all operators which are Members or Associate Members of IATA;
- all operators which are party to the IATA Multilateral Interline Traffic Agreement—Cargo; and
- all shippers and agents that offer consignments of dangerous goods to these operators.

1.2.2 Relationship to ICAO

1.2.2.1 The ICAO Annex 18 and the *Technical Instructions for the Safe Transport of Dangerous Goods by Air* are applicable for the transport of dangerous goods by air from, to or through the Member States of ICAO.

1.2.2.2 There are certain differences between the IATA and ICAO regulations which stem from operational considerations and result in a regulatory regime which is necessarily more restrictive than the ICAO requirements. Differences which are more restrictive than the ICAO *Technical Instructions* are identified in the Regulations by the symbol “” appearing in the margin. The IATA *Dangerous Goods Regulations* also incorporates additional material of practical assistance to users.

1.2.3 General

1.2.3.1 These Dangerous Goods Regulations, referred to as the "Regulations", prescribe the detailed requirements applicable to the international transport of dangerous goods by air under normal circumstances. Any addendum to this edition of the Dangerous Goods Regulations issued by IATA constitutes part of the Regulations.

1.2.3.2 In these Regulations, the words "shall" and "must" are used to indicate a mandatory requirement. The words "should" and "may" indicate a preferred requirement and are not binding.

1.2.4 Scope

Nothing contained in these Regulations should be interpreted as:

- (a) requiring an operator to transport a particular article or substance;
- (b) preventing an operator from imposing special requirements on the transport of a particular article or substance over and above the requirements contained herein; or
- (c) preventing an operator from requiring a shipper to seek confirmation or endorsement of the "Shipper's Declaration for Dangerous Goods" from an authority named by the operator.

1.2.5 Approvals

STATE VARIATIONS: AUG-01, BEG-02/04, BHG-02/03, BRG-11, CAG-07/08, FRG-01, GHG-02/03, HRG-03, IDG-01, ITG-05, IRG-03/04, JMG-01, KPG-02, MOG-02, NLG-01, OMG-03, ROG-01/02, USG-03, VEG-01, YEG-03/06, ZAG-01

OPERATOR VARIATIONS: 2K-12, 4C-01, 5X-08/09, 6R-13, 7H-01, AF-03, AM-15, AV-12, AY-03, BZ-09, CI-07, D0-05, D5-04, EK-02, ES-05, EX-12, FX-04, GU-12, JJ-01, KL-03, L7-01, LA-01, LP-01, LR-12, LU-01, M3-01, MP-03, P3-06, PZ-01, QR-07, QT-13, QY-05, RU-06, SV-14, TA-12, TG-07, TN-02, UC-01, V8-06, WC-12, XL-01

1.2.5.1 Where specifically provided for in these Regulations, the States concerned may grant an approval to permit the transport of dangerous goods, provided that in such instances an overall level of safety in transport which is equivalent to the level of safety provided for in these Regulations is achieved.

Note:

For the purposes of approvals, "States concerned" are the States of origin and the operator, unless otherwise specified in these Regulations.

1.2.5.2 Acceptance of dangerous goods offered for transport under the provisions of an approval is at the discretion of the operator(s) concerned. Shippers are encouraged to make advance arrangements with the operator(s) as part of the planning process associated with any approval application.

1.2.6 Exemptions

STATE VARIATIONS: BRG-04, DEG-04, ROG-03, YEG-03

OPERATOR VARIATIONS: 2K-12, 4C-01, 6R-13, AF-03, AM-15, AV-12, AY-03, BZ-08/09, CI-07, EK-02, EX-12, GA-11, GU-12, JJ-01, KL-03, L7-01, LA-01, LP-01, LR-12, LU-01, M3-01, MP-03, P3-06, PZ-01, QT-13, RU-06, TA-12, UC-01, V8-06, XL-01

1.2.6.1 In instances of extreme urgency or when other forms of transport are inappropriate or when full compliance with the prescribed requirements is contrary to the public interest, the States concerned may grant exemption from the provisions of the Regulations provided that in such instances every effort is made to achieve an overall level of safety in transport which is equivalent to the level of safety provided for in these Regulations.

Note:

Refer to 2.1.1 for dangerous goods forbidden for transport by air under any circumstances.

1.2.6.2 For the State of overflight, if none of the criteria for granting an exemption are relevant, an exemption may be granted based solely on whether it is believed that an equivalent level of safety in air transport has been achieved.

Notes:

1. *For the purposes of exemptions, the "States concerned" are the States of origin, operator, transit, overflight and destination.*
2. *Guidance for the processing of exemptions, including examples of extreme urgency, may be found in the ICAO Supplement to the Technical Instructions (Part S-1; 1.2 and 1.3).*
3. *Due to differences in the type of operations carried out by helicopters compared with aeroplanes, some additional considerations need to be made when dangerous goods are carried by helicopter, as described in Subsection 9.9.*

1.2.6.3 The exemption should include, as a minimum, the following:

- (a) the UN/ID number, proper shipping name and classification;
- (b) packaging and quantity applicable;
- (c) any special handling required and any special emergency response information;
- (d) name and address of shipper and consignee;
- (e) airports of departure, transit and destination and the proposed dates of transport; and
- (f) duration of validity of the exemption.

1.2.6.4 A copy of the exemption issued by all States concerned must be provided to the operator and must accompany the consignment. If the exemption documents are not in English, an accurate translation in English must accompany the consignment.

1.2.6.5 Acceptance of dangerous goods offered for transport under the provisions of an exemption is at the discretion of the operator(s) concerned. Shippers are encouraged to make advance arrangements with the

operator(s) as part of the planning process associated with any exemption application.

1.2.7 Exceptions

1.2.7.1 Except for information provided to operator employees, as shown in 1.4.2, the provisions of these Regulations do not apply to dangerous goods carried by an aircraft where the dangerous goods are:

- (a) to provide, during flight, medical aid to a patient or to preserve organs intended for use in transplantation when those dangerous goods:
 1. have been placed on board with the approval of the operator; or
 2. form part of the permanent equipment of the aircraft when it has been adapted for specialized use; providing that:
 - (i) gas cylinders have been manufactured specifically for the purpose of containing and transporting that particular gas;
 - (ii) equipment containing wet cell batteries is kept and when necessary secured, in an upright position to prevent spillage of the electrolyte;
 - (iii) lithium metal or lithium ion cells or batteries meet the provisions of 3.9.2.6.1. Spare lithium batteries, when not in use, must be individually protected so as to prevent short circuits.

Note:

For the dangerous goods passengers are permitted to carry as medical aid see 2.3.2 to 2.3.5.

- (b) to provide veterinary aid or a humane killer for an animal during flight;
- (c) for dropping in connection with agricultural, horticultural, forestry, ice jam control, landslide clearance, pollution control or pest management activities;
- (d) for dropping or triggering in connection with avalanche control activities;
- (e) to provide aid in connection with search and rescue operations during flight, or related to the flight;
- (f) vehicles carried in aircraft designed or modified for vehicle ferry operations if all of the following requirements are met:
 1. authorization has been given by the appropriate authorities of the States concerned and such authorities have prescribed specific terms and conditions for the particular operator's operation;
 2. vehicles are secured in an upright position;
 3. fuel tanks are so filled as to prevent spillage of fuel during loading, unloading and transit; and
 4. adequate ventilation rates are maintained in the aircraft compartment in which the vehicles are carried.
- (g) dangerous goods that are required for the propulsion of the means of transport or the operation of its specialized equipment during transport (e.g. refrigeration units) or that are required in accordance with the operating regulations (e.g. fire extinguishers) (see Subsection 2.5).

Note:

This exception is only applicable to the means of transport performing the transport operation.

- (h) contained within items of excess baggage (see definition of "excess baggage" in Appendix A—Glossary) being sent as cargo provided that:
 1. the excess baggage has been consigned as cargo by or on behalf of a passenger;
 2. the dangerous goods may only be those that are permitted by and in accordance with 2.3 to be carried in checked baggage;
 3. the excess baggage is marked with the words "Excess baggage consigned as cargo".

1.2.7.2 Provision must be made to stow and secure dangerous goods transported under 1.2.7.1(a), (b), (c) and (d) during take-off and landing and at all other times when deemed necessary by the pilot-in-command.

1.2.7.3 The dangerous goods must be under the control of trained personnel during the time when they are in use on the aircraft.

1.2.7.4 Dangerous goods transported under 1.2.7.1(a), (b), (c), (d) and (e) may be carried on a flight made by the same aircraft before or after a flight for the purposes identified above, when it is impracticable to load or unload the dangerous goods immediately before or after the flight, subject to the following conditions:

- (a) the dangerous goods must be capable of withstanding the normal conditions of air transport;
- (b) the dangerous goods must be appropriately identified (e.g. by marking or labelling);
- (c) the dangerous goods may only be carried with the approval of the operator;
- (d) the dangerous goods must be inspected for damage or leakage prior to loading;
- (e) loading must be supervised by the operator;
- (f) the dangerous goods must be stowed and secured in the aircraft in a manner that will prevent any movement in flight which would change their orientation;
- (g) the pilot-in-command must be notified of the dangerous goods loaded onboard the aircraft and their loading location. In the event of a crew change, this information must be passed to the next crew;
- (h) all personnel must be trained commensurate with the functions for which they are responsible;
- (i) the provisions of 1.4.2 and 9.6.1 apply.

1.2.7.5 Dangerous goods transported under 1.2.7.1(a), (b), (c), (d) and (e) may be carried on flights made by the same aircraft for other purposes (e.g. training flights and positioning flights prior to or after maintenance), subject to the conditions in 1.2.7.4(a) to (i).

1.2.8 General Transport Requirements

STATE VARIATIONS: CNG-01, HKG-01, KHG-01, ING-01, MYG-01, NLG-06, SGG-01, VCG-01/03

1.2.8.1 Except as otherwise provided for in these Regulations, no person may offer or accept dangerous goods for air transport unless those goods are properly classified, documented, certificated, described, packaged,

1

1.2
to
1.3

marked, labelled and in the condition for shipment required by these Regulations. If a person performs a function required by these Regulations on behalf of the person who offers the dangerous goods for transport by air or on behalf of the operator, that person must perform that function in accordance with the requirements of these Regulations. No person may transport dangerous goods by air unless those goods are accepted, handled and transported in accordance with these Regulations. No person may label, mark, certify or offer a packaging as meeting the requirements of these Regulations unless that packaging is manufactured, fabricated, marked, maintained, reconditioned or repaired as required by these Regulations. No person shall carry dangerous goods or cause dangerous goods to be carried aboard an aircraft in either checked or carry-on baggage or on their person, unless permitted by Subsection 2.3.

Note:

When dangerous goods intended for air transport are carried by surface transport to or from an airport, any other applicable national or modal transport requirements should also be met in addition to those that are applicable for the goods when carried by air.

1.2.8.2 Consignments of dangerous goods accepted for air transport which meet all of the requirements of the most current edition of these Regulations, may continue on the same journey, including transfers to the operator of any subsequent aircraft used as part of the same journey, even though an addendum has been issued to the Regulations, or a new edition of these Regulations has come into effect.

1.2.9 Application of Standards

Where the application of a standard is required and there is any conflict between the standard and these Regulations, the Regulations take precedence. The requirements of the standard that do not conflict with these Regulations must be applied as specified, including the requirements of any other standard, or part of a standard, referenced within that standard as normative.

1.2.10 Dangerous Goods Packages Opened by Customs and Other Authorities

Any package opened during an inspection must, before being forwarded to the consignee, be restored by qualified persons to a condition, which complies with these Regulations.

1.2.11 Lamps Containing Dangerous Goods

1.2.11.1 The following lamps are not subject to these Regulations provided that they do not contain radioactive material:

- (a) lamps each containing not more than 1 g of dangerous goods and packaged so that there is not more than 30 g of dangerous goods per package, provided that:
 1. the lamps are certified to a manufacturer's quality management system; and

Note:

The application of ISO 9001:2008 may be considered acceptable for this purpose.

2. each lamp is either individually packed in inner packagings, separated by dividers, or surrounded with cushioning material to protect the lamps and packed into strong outer packagings meeting the general provisions of 5.0.2.4.1 and capable of passing a 1.2 m drop test.
- (b) lamps containing only gases of Division 2.2 (according to 3.2.2.2) provided they are packaged so that the projectile effects of any rupture of the bulb will be contained within the package.

Note:

Lamps containing radioactive material are addressed in 10.3.2.2(b).

1.3 Shipper's Responsibilities

1.3.1 Compliance

1.3.1.1 A shipper must comply fully with these Regulations when offering a consignment of dangerous goods to IATA Member and associate Member airlines and to airlines participating in IATA interline agreements for cargo. In addition, shippers must comply with any applicable regulations set forth by the States of origin, transit and destination.

1.3.1.2 These Regulations are fully compliant with the ICAO *Technical Instructions*. A shipper, offering articles or substances in violation of these Regulations, may be in breach of national law and may be subject to legal penalties.

1.3.1.3 It is the shipper's responsibility to ensure that all of the applicable air transport requirements are met. The items indicated in 1.3.2 are provided as examples and do not include the complete list of all the applicable requirements for air transport.

1.3.2 Specific Responsibilities

STATE VARIATIONS: BEG-04, BHG-01, BRG-02, CAG-06, ETG-01, NLG-03, SAG-02, USG-04

OPERATOR VARIATIONS: 5X-01/03/04, 6O-01, AY-01, D0-01, ES-01/05, ET-01, G3-01, GA-01, GK-01, HM-05, IY-01, JU-01, KC-02, KE-02, KZ-01, ME-03, MH-01, MK-06, MS-02, NH-01, O3-02, OM-01, OU-01, OZ-01, QY-01, PG-01, RJ-01, RO-02, S7-01, SN-10, TK-04, UU-04/07, VN-01, WN-01, WY-01/05/10

Before any package or overpack of dangerous goods is offered for air transport, the shipper must comply with the following specific responsibilities:

- (a) a shipper must provide such information to their employees as will enable them to carry out the functions for which they are responsible with regard to the transport of dangerous goods by air;
- (b) the shipper must ensure that the articles or substances are not prohibited for transport by air (see Subsection 2.1 and Subsection 4.2);

- (c) the articles or substances must be properly identified, classified, packed, marked, labelled, documented and be in the condition for transport in accordance with these Regulations;
- (d) before a consignment of dangerous goods is offered for air transport, all relevant persons involved in its preparation must have received training to enable them to carry out the functions for which they are responsible, as detailed in Subsection 1.5. Where a shipper does not have trained staff, the "relevant persons" may be interpreted as applying to those employed to act on the shipper's behalf and undertake the shipper's responsibilities in the preparation of the consignment. However, such persons must be trained as required by Subsection 1.5;
- (e) the dangerous goods are packed in compliance with all applicable air transport requirements including:
 - inner packaging and the maximum quantity per package limits;
 - appropriate types of packaging according to the packing instructions;
 - other applicable requirements indicated in the packing instructions including:
 - single packagings may be forbidden;
 - only inner and outer packagings indicated in the packing instructions are permitted;
 - inner packaging may need to be packed in intermediate packagings; and
 - certain dangerous goods must be transported in packagings meeting a higher performance level.
 - appropriate closure procedures for inner and outer packagings (see 5.0.2.7);
 - the compatibility requirements as specified in the applicable packing instructions and in 5.0.2;
 - the absorbent materials requirements in the packing instructions when applicable; and
 - the pressure differential requirement of 5.0.2.9.

1.3.3 Dangerous Goods in Consolidations

1.3.3.1 Definition

A consolidation is a consignment of multi-packages which has been originated by more than one person each of whom has made an agreement for carriage by air with another person other than a scheduled air carrier. Conditions applied to that agreement may or may not be the same as conditions applied by the scheduled air carrier for the same carriage.

1.3.3.2 Requirements

Dangerous goods are accepted in consolidations under the conditions specified in 1.3.3.2.1 to 1.3.3.2.6.

1.3.3.2.1 Dangerous goods may be consolidated with goods not subject to these Regulations. Dangerous goods in consolidations are subject to the acceptance check described in 9.1.3. Any delays caused by discrepancies found during the check may result in delay to the complete consolidation.

1.3.3.2.2 Dangerous goods in consolidations must be identified, classified, packed, marked, labelled and documented in accordance with these Regulations and be free from any indication of damage or leakage.

1.3.3.2.3 Packages and overpacks containing dangerous goods must be offered to the operator separately from the goods in the consolidation that are not subject to these Regulations. Dangerous goods in consolidations are not acceptable in unit load devices, unless specifically permitted by these Regulations (see 9.1.4.1).

1.3.3.2.4 A Shipper's Declaration for Dangerous Goods is required for each component consignment.

1.3.3.2.5 Consolidations containing any "Cargo Aircraft Only" dangerous goods must be shipped on Cargo Aircraft.

1.3.3.2.6 Before a consignment is offered to an operator for transport, the shipper, the freight forwarder and the cargo agent must:

- ensure that the dangerous goods are in full compliance with the Regulations;
- segregate dangerous goods contained in a consolidation from goods which are not subject to the Regulations and offer them separately;
- ensure that the dangerous goods are not loaded in a unit load device other than those permitted under 9.1.4.1;
- for all consignments, check documents and the exterior of packages for indication of hidden hazards.

1.3.4 Retention of Documents

STATE VARIATION: USG-01

1.3.4.1 The shipper must ensure that at least one copy of the documents, appropriate to the transport by air of a dangerous goods consignment, is retained for a minimum period of three months. As a minimum, the documents which must be retained are the Shipper's Declaration for Dangerous Goods and any other transport documents applicable to the consignment as specified in these Regulations.

1.3.4.2 Where the documents are kept electronically or in a computer system, the shipper must be able to reproduce them in a printed form.

1.4 Operator's Responsibilities

STATE VARIATIONS: NLG-06, USG-13

1.4.1 General

In transporting dangerous goods, an operator must comply with the requirements of Section 9 for:

- Acceptance;
- Storage;
- Loading;
- Inspection;
- Provision of Information, including emergency response information;
- Reporting;
- Retention of Records;
- Training.

Note:

When an operator, its subsidiary or an agent of the operator offers a consignment of dangerous goods for air transport then the operator, its subsidiary or the agent is a shipper and must comply with shipper's responsibilities (see Subsection 1.3). This is applicable even if the consignment is to be transported on its own or on other operator's services.

1.4.2 Information to Operator Employees

1.4.2.1 An operator must provide, in the operator's operations and/or other appropriate manuals, information to employees so as to enable flight crews and other employees to carry out the functions for which they are responsible with regard to dangerous goods. Where applicable, this information must also be provided to ground handling agents. This information must include:

- (a) for passenger handling staff and cabin crew the procedures to be followed to alert passengers that certain items of dangerous goods are specifically prohibited from being in checked baggage, e.g. spare lithium batteries (see Subsection 2.3) and must be removed from baggage where items of carry-on baggage cannot be accommodated in the cabin;
- (b) the action to be taken in the event of emergencies involving dangerous goods;
- (c) details of the location and identification of cargo compartments;
- (d) the maximum quantity of dry ice permitted in each compartment; and
- (e) if radioactive material is to be carried, instructions on the loading of such dangerous goods, based on the requirements of 10.9.3.

1.4.2.2 In addition to the above, it is recommended that the operator's operations and/or other appropriate manuals should contain information specific to dangerous goods permitted in passenger and crew baggage as permitted by Subsection 2.3. The information in the operator's manuals should address:

- (a) **approval process.** It is recommended that a single company policy be set out that identifies the items that have been approved and the person(s) or department(s) responsible for determining how dangerous goods in passenger baggage may be approved;
- (b) **communication.** It is recommended that the operator define how approvals for dangerous goods requiring operator approval are communicated to the airport(s) of departure. It is recommended that operators consider a process where such approval is included in the passenger(s) electronic record;
- (c) **limitations.** The operator manuals should specify any limitations or procedural requirements that may apply to particular commodities, e.g. inspection at check-in by passenger service agents and/or security;
- (d) **interlining.** Where the operator has interline agreements with code share and/or alliance partners the operator should identify what the procedure is for obtaining the approval of the other operator(s) involved, e.g. by advising the passenger that they must obtain approval from the other operator;
- (e) **awareness.** The operator should ensure that all staff who have an interaction with passengers, (i.e. reservations agents, passenger service agents, cabin crew and flight crew) are made aware of the process employed to ensure that the operator approval process remains effective.
- (f) **information to the pilot-in-command.** Information must be provided to the pilot-in-command when certain dangerous goods permitted in passenger or crew baggage are accepted for carriage. It is recommended that operators implement and document the following procedures to address pilot notification:
 1. passenger handling staff communicate information to load planning to advise:
 - (i) the detail of the item accepted, e.g. wheelchair with lithium ion battery, oxygen, etc.;
 - (ii) the name of the passenger and seat number; and
 - (iii) if the item is being carried in the cabin or if it will be loaded in the cargo compartment.
 2. load planning staff should:
 - (i) insert a free text supplementary information (SI) item on the loadsheet to show the information indicated in (f)1.(i) and (ii) and for items loaded in the cargo compartment the loading location;
 - (ii) communicate to ramp the required loading location for items loaded in the cargo compartment.

Note:

The requirements for the notification to pilot-in-command as shown in 9.5.1.1.3 do not apply to items being carried by passengers as either checked or carry-on baggage.

1.4.3 Provision of Information to Passengers

1.4.3.1 Operators must inform passengers about dangerous goods that passengers are forbidden to carry aboard an aircraft. The notification system must be described in their operations manual and/or other appropriate manuals. The notification system must ensure that where the ticket purchase and/or boarding pass issuance can be completed by a passenger without the involvement of another person, the system must include an acknowledgement by the passenger that they have been presented with the information. The information must be provided to passengers:

- (a) at the point of ticket purchase or, if this is not practical, made available in another manner to passengers prior to boarding pass issuance; and
- (b) at boarding pass issuance, or when no boarding pass is issued, prior to boarding the aircraft.

Note:

The information may be provided in text or pictorial form, electronically, or verbally, as described in the operator's manuals.

1.4.3.2 An operator or the operator's handling agent and the airport operator must ensure that information on the types of dangerous goods which are forbidden for carriage aboard an aircraft is communicated effectively to passengers. This information must be presented:

- (a) at each of the places at an airport:
 - where tickets are issued;
 - boarding passes are issued;
 - passenger baggage is dropped off; and
 - in aircraft boarding areas.
- (b) any other location where:
 - 1. passengers are issued boarding passes; and
 - 2. checked baggage is accepted.
- (c) should be prominently displayed in sufficient numbers in baggage claim areas.

1.4.3.2.1 This information must include visual examples of dangerous goods forbidden from carriage aboard an aircraft.

1.4.3.3 An operator, of passenger aircraft, should have information on those dangerous goods which may be carried by passengers in accordance with 2.3.2 to 2.3.5 available prior to the boarding pass issuance process on their web sites or other sources of information.

1.4.3.4 Any organization or enterprise other than an operator (such as a travel agent) involved in the air transport of passengers, should provide passengers with information about the types of dangerous goods which they are forbidden from carrying aboard an aircraft. This information should consist as a minimum of notices at those locations where there is an interface with the passengers.

1.4.4 Passenger Check-in Procedures

1.4.4.1 Operators' check-in staff must be adequately trained to assist them to identify and detect dangerous goods carried by passengers other than as permitted in Subsection 2.3.

1.4.4.2 Many innocuous-looking items may contain dangerous goods and a list of general descriptions which, experience has shown, are often applied to such items is shown in Subsection 2.2. Check-in staff should therefore seek confirmation from any passenger where there are suspicions that an item of baggage may contain dangerous goods that are not permitted.

1.4.5 Provision of Information at Cargo Acceptance Areas

An operator or the operator's handling agent must ensure that sufficient notices, prominently displayed, are provided at visible location(s) at cargo acceptance points, giving information about the transport of dangerous goods to alert shippers/agents about any dangerous goods that may be contained in their cargo consignment(s). These notices must include visual examples of dangerous goods, including batteries.

1.5 Training Requirements

STATE VARIATIONS: AEG-02, CAG-11, GBG-07, HKG-01, NLG-06, VEG-05

1.5.1 Dangerous Goods Training Programs

1.5.1.1 Establishment and Maintenance

Note:

A training program includes elements such as design methodology, assessment, initial and recurrent training, instructor qualifications and competencies, training records and evaluation of the effectiveness of training.

1.5.1.1.1 The employer of personnel that perform functions aimed at ensuring that dangerous goods are transported in accordance with these Regulations must establish and maintain a dangerous goods training program.

Notes:

- △ 1. An approach to ensuring personnel are competent to perform any function for which they are responsible is provided in "DANGEROUS GOODS TRAINING PROGRAMS GUIDELINES - COMPETENCY-BASED TRAINING APPROACH", which is available on: www.iata.org/dangerousgoods
- 2. Security personnel who are involved with the screening of passengers and crew and their baggage and cargo or mail are required to be trained irrespective of whether the operator on which passenger or cargo is to be transported carries dangerous goods as cargo.

1.5.1.1.2 All operators must establish a dangerous goods training program regardless of whether or not they are approved to transport dangerous goods as cargo.

1.5.1.1.3 Training courses may be developed and delivered by or for the employer.

1.5.1.2 Objective of Dangerous Goods Training

1.5.1.2.1 The employer must ensure that personnel are competent to perform any function for which they are responsible prior to performing any of these functions. This must be achieved through training and assessment commensurate with the functions for which they are responsible. Such training must include:

- general familiarization training—personnel must be trained to be familiar with the general provisions;
- function specific training—personnel must be trained to competently perform the function for which they are responsible; and
- safety training—personnel must be trained on how to recognise the hazards presented by dangerous goods, on the safe handling of dangerous goods and on emergency response procedures.

Note:

General information on the provisions for dangerous goods carried by passengers and crew (see 2.3) should be included in training courses, as appropriate.

1.5.1.2.2 Personnel who have received training but who are assigned to new functions must be assessed to determine their competence in respect of their new function. If competency is not demonstrated, appropriate additional training must be provided.

1.5.1.3 Recurrent Training and Assessment

Personnel must receive recurrent training and assessment within 24 months of previous training and assessment to ensure that competency has been maintained. However, if recurrent training and assessment is completed within the final three months of validity of the previous training and assessment, the period of validity extends from the month on which the recurrent training and assessment was completed until 24 months from the expiry month of that previous training and assessment. For example, a person attends an initial course finishing on 14 April 2019; their training validity therefore expires 30 April 2021. They may attend recurrent training any time between 1 February and 30 April 2021 and their next recurrent training date will remain 30 April 2023. If, however they complete recurrent training in January 2021, then as this is more than 3 months prior to the end of April then their training expiry date becomes 31 January 2023.

1.5.1.4 Training and Assessment Records

STATE VARIATIONS: CAG-11, VEG-05

1.5.1.4.1 The employer must maintain a record of training and assessment for personnel.

1.5.1.4.2 The record of training and assessment must be maintained, which must include:

- the individual's name;
- the most recent training and assessment completion month;
- a description, copy or reference to training and assessment materials used to meet the training and assessment requirements;
- the name and address of the organization providing the training and assessment; and
- evidence which shows that personnel have been assessed as competent to perform any function for which they are responsible.

1.5.1.4.3 Training and assessment records must be retained by the employer for a minimum period of 36 months from the most recent training and assessment completion month and must be made available upon request to the employee or appropriate national authority.

1.5.2 Review and Approval of Training Programs

1.5.2.1 Dangerous goods training programs for operators' must be approved by the appropriate authority of the State of the operator in accordance with the provisions of ICAO Annex 6—Operation of Aircraft.

1.5.2.2 Dangerous goods training programs required for entities other than operators and designated postal operators should be approved as determined by the appropriate national authority.

Notes:

- See 1.5.4.2 for approval of training programs for designated postal operators.
- See 1.5.5 for information on detailed dangerous goods training course content developed by IATA that address well-defined job functions.

1.5.3 Instructor Qualifications and Competencies

1.5.3.1 Unless otherwise provided for by the appropriate national authority, instructors of initial and recurrent dangerous goods training programs must demonstrate or be assessed as competent in instruction and the function(s) that they will instruct prior to delivering such a dangerous goods training program.

Note:

"Competence in instruction" can come from a variety of methods. A list of instructional techniques is found in the IATA Dangerous Goods Regulations Instructors Toolkit. See <https://www.iata.org/en/publications/store/dgr-training-books/>

1.5.3.2 Instructors delivering initial and recurrent dangerous goods training programs must deliver such a course at least every 24 months, or in the absence of this attend recurrent training.

1.5.3.3 Organisations must ensure that the instructor receives updates to the Regulations and training material on an annual basis with the issuance of each edition of the DGR or as the Regulations are modified. Instructors must receive and understand updates to dangerous goods information and be made familiar with those changes by training or other means on an annual basis or as the Regulations are modified.

1.5.4 Designated Postal Operators Training Programs

1.5.4.1 Staff of designated postal operators must be trained commensurate with their responsibilities. The subject matter to which their various categories of staff should be familiar with is indicated in Table 1.5.A.

1.5.4.2 Dangerous goods training programs of designated postal operators must be subjected to review and approval by the civil aviation authority of the State where the mail was accepted by the designated postal operator.

TABLE 1.5.A
Minimum Requirements for Training Curricula for Designated Postal Operators (1.5.1.2)

Aspects of transport of dangerous goods by air with which they should be familiar, as a minimum	Designated Postal Operators		
	Category		
	a	b	c
General philosophy	X	X	X
Limitations	X	X	X
General requirements for shippers	X		
Classification	X		
List of dangerous goods	X		
General packing requirements	X		
Packing instructions	X		
Labelling and marking	X	X	X
Shipper's Declaration and other relevant documentation	X	X	
Acceptance of the dangerous goods listed in 2.4	X		
Recognition of undeclared dangerous goods	X	X	X
Storage and loading procedures			X
Provisions for passengers and crew	X	X	X
Emergency procedures	X	X	X

CATEGORY

- (a) Staff of designated postal operators involved in accepting mail containing dangerous goods
- (b) Staff of designated postal operators involved in processing mail (other than dangerous goods)
- (c) Staff of designated postal operators involved in the handling, storage and loading of mail

1.5.5 IATA Dangerous Goods Training Workbooks

IATA has developed a series of training workbooks that correspond to well-defined job functions that share similar competencies:

- Workbook 1—Shippers; Packers; Personnel Accepting/Processing Dangerous Goods
- Workbook 2—Flight Crew; Load Planners; Flight Dispatchers
- Workbook 3—Cabin Crew; Passenger Handling Personnel; Security Screening Personnel
- Workbook 4—Ramp and Warehouse Personnel
- Workbook 5—General Cargo Acceptance Personnel

For more information see:

<https://www.iata.org/en/publications/store/dgr-training-books/>

1.6 Adequate Instruction for Shipping Section II Lithium Batteries

1.6.1 Section II of the lithium battery packing instructions, PI 966, PI 967, PI 969 and PI 970, include a requirement that “Any person preparing or offering cells or batteries for transport must receive adequate instruction on these requirements commensurate with the functions for which they are responsible”. The packing instructions however do not define or describe what is considered as “adequate instruction”.

1.6.2 At a minimum, an employer should consider the following as being adequate instruction:

- Classification of lithium batteries being shipped.
- Documentation of procedures applied to lithium batteries being shipped.
- Written work instructions or other documentation, including automated controls.
- Review and understanding of documented procedures as applicable to the job function.
- Instruction records including date(s) for all employees.
- Refresher instructions provided at a minimum every two years or as the documented instructions are revised or regulations are changed.
- Reverse logistics, including transport mode and applicable prohibitions.

1.7 Dangerous Goods Security

STATE VARIATION: USG-17

1.7.0 General

1.7.0.1 This subsection addresses the security responsibilities of operators, shippers and others involved in the transport of dangerous goods aboard aircraft. It should be noted that ICAO Annex 17—Security provides comprehensive requirements for implementation of security measures by States to prevent unlawful interference with civil aviation or when such interference has been

committed. In addition, the ICAO *Security Manual for Safeguarding Civil Aviation against Acts of Unlawful Interference (Doc 8973 - Restricted)* provides procedures and guidance on aspects of aviation security and is intended to assist States in the implementation of their respective national civil aviation security programs. The current edition of the IATA Security Manual contains guidance material directed at industry entities such as operators and airports. The requirements in this subsection are intended to supplement the requirements of Annex 17 and to implement measures to be taken to minimize theft or misuse of dangerous goods that may endanger persons or property. The provisions of this subsection do not supersede the requirements of Annex 17 and mandatory elements of the associated documents.

1.7.0.2 In addition to the security provisions of these Regulations, appropriate national authorities may implement further security provisions for reasons other than safety of dangerous goods during transport. In order to not impede international and multimodal transport by different explosives security marks, it is recommended that such marks be formatted consistent with an internationally harmonized standard (e.g. European Union Commission Directive 2008/43/EC).

1.7.1 General Security Provisions

1.7.1.1 All persons engaged in the transport of dangerous goods should consider security requirements for the dangerous goods commensurate with their responsibilities.

1.7.1.2 Dangerous goods should only be offered to operators that have been appropriately identified.

1.7.1.3 The provisions of this subsection do not apply to:

- (a) UN 2908 and UN 2909 excepted packages;
- (b) UN 2910 and UN 2911 excepted packages with an activity level not exceeding the A₂ value; and
- (c) UN 2912 LSA-I and UN 2913 SCO-I.

1.7.2 Dangerous Goods Security Training

1.7.2.1 The training specified in Subsection 1.5 should include elements of security awareness.

1.7.2.2 Security awareness training should address the nature of security risks, recognition of security risks, methods to address and reduce such risks and actions to be taken in the event of a security breach. It should include awareness of security plans (if appropriate) commensurate with the responsibilities of individuals and their role in implementing security plans.

Note:

Persons who have received security training in accordance with the requirements of a National Security Plan or other security requirements that fulfil the elements of 1.7.2.2 need not receive additional training.

1.7.2.3 Such training should be provided or verified upon employment in a position involving dangerous goods transport. Recurrent training should take place

within 24 months of previous training to ensure knowledge is current.

1.7.2.4 Records of all dangerous goods security training undertaken should be kept by the employer and made available to the employee or appropriate national authority if requested. Records should be kept by the employer for a period of time established by the appropriate national authority.

1.7.3 Provisions for High Consequence Dangerous Goods

OPERATOR VARIATIONS: BZ-07, LO-01, TK-01, VA-02

1.7.3.1 Definition of High Consequence Dangerous Goods

1.7.3.1.1 High consequence dangerous goods are those which have the potential for misuse in a terrorist event and which may, as a result, produce serious consequences such as mass casualties, mass destruction or, particularly for Class 7, mass socio-economic disruption.

1.7.3.1.2 An indicative list of high consequence dangerous goods in classes and divisions other than Class 7 is given in Table 1.7.A.

TABLE 1.7.A
Indicative List of High Consequence Dangerous Goods (1.7.3.1.2)

Class 1	Division 1.1
Class 1	Division 1.2
Class 1	Division 1.3 compatibility group C
Class 1	Division 1.4, UN 0104, UN 0237, UN 0255, UN 0267, UN 0289, UN 0361, UN 0365, UN 0366, UN 0440, UN 0441, UN 0455, UN 0456, UN 0500, UN 0512, UN 0513
Class 1	Division 1.5
Class 1	Division 1.6
Class 2	Division 2.3 toxic gases (excluding aerosols)
Class 3	desensitized explosives
Class 4	Division 4.1 desensitized explosives
Class 6	Division 6.1 substances of Packing Group I; except when transported under the excepted quantity provisions (see 2.6)
Class 6	Division 6.2 infectious substances of Category A (UN 2814 and UN 2900) and medical waste of Category A (UN 3549)

1.7.3.1.3 For dangerous goods of Class 7, high consequence radioactive material is that with an activity equal to or greater than a transport security threshold of 3,000 A₂ per single package (see also 10.3.2.1) except for the following radionuclides where the transport security threshold is given in Table 1.7.B.

1.7.3.1.4 For mixtures of radionuclides, determination of whether or not the transport security threshold has been met or exceeded can be calculated by summing the ratios of activity present for each radionuclide divided by the transport security threshold for that radionuclide. If the sum of the fractions is less than 1, then the radioactivity threshold for the mixture has not been met nor exceeded.

This calculation can be made with the formula:

$$\sum \frac{A_i}{T_i} < 1$$

Where:

A_i = activity of radionuclide i that is present in a package (TBq)

T_i = transport security threshold for radionuclide i (TBq).

1.7.3.1.5 When radioactive material possesses subsidiary hazards of other classes or divisions, the criteria of Table 1.7.A should also be taken into account (see also 10.0.5).

TABLE 1.7.B
Transport Security Thresholds for Specific Radionuclides (1.7.3.1.3)

Radionuclide	Element	Transport Security Threshold (TBq)
Am-241	Americium	0.6
Au-198	Gold	2
Cd-109	Cadmium	200
Cf-252	Californium	0.2
Cm-244	Curium	0.5
Co-57	Cobalt	7
Co-60		0.3
Cs-137	Caesium	1
Fe-55	Iron	8,000
Gd-153	Gadolinium	10
Ge-68	Germanium	7
Ir-192	Iridium	0.8
Ni-63	Nickel	600
Pd-103	Palladium	900
Pm-147	Promethium	400
Po-210	Polonium	0.6
Pu-238	Plutonium	0.6
Pu-239		0.6
Ra-226	Radium	0.4
Ru-106	Ruthenium	3
Se-75	Selenium	2
Sr-90	Strontium	10
Tl-204	Thallium	200
Tm-170	Thulium	200
Yb-169	Ytterbium	3

1.7.4 Security Plans

1.7.4.1 Applicability

Operators, shippers and others (including infrastructure managers) engaged in the transport of high consequence dangerous goods (see 1.7.3) should adopt, implement and comply with a security plan that addresses at least the elements specified in 1.7.4.2.

Note:

When national authorities issue exemptions, they should consider all of the provisions in this Section.

1.7.4.2 Elements of a Security Plan

At the minimum, the security plan should comprise of the following elements:

- specific allocation of responsibilities for security to competent and qualified persons with appropriate authority to carry out their responsibilities;
- records of dangerous goods or types of dangerous goods transported;
- review of current operations and assessment of vulnerabilities, including inter-modal transfer, temporary transit storage, handling and distribution as appropriate;
- clear statement of measures including training policies (including response to higher threat conditions, new employee/employment verifications etc.), operating practices (e.g. access to dangerous goods in temporary storage proximity to vulnerable infrastructure etc.), equipment and resources that are to be used to reduce security risks;
- effective and up to date procedures for reporting and dealing with security threats, breaches of security or security incidents;
- procedures for the evaluation and testing of security plans and procedures for periodic review and update of the plans;
- measures to ensure the security of transport information contained in the plan; and
- measures to ensure that the security of the distribution of transport documentation is limited as far as possible (such measures must not preclude provision of the transport documentation required by Section 8 of these Regulations).

Note:

Operators, shippers and others with responsibilities for the safe and secure transport of dangerous goods should cooperate with each other and with appropriate authorities to exchange threat information, apply appropriate security measures and respond to security incidents.

△ 1.7.5 Radioactive Material

For radioactive material, the provisions of this Subsection are deemed to be complied with when the provisions of the Convention on Physical Protection of Nuclear Material (INFCIRC/274/Rev.1. IAEA Vienna (1980)) and of IAEA circular on "Nuclear Security Recommendations on Physical Protection of Nuclear Material and Nuclear Facilities" (INFCIRC/225/Rev.5. IAEA Vienna (2011)) are applied.

1.8 Incident and Accident Reporting

Entities other than operators who are in possession of dangerous goods at the time a dangerous goods accident or incident occurs or at the time a dangerous goods incident is discovered to have occurred should follow the reporting requirements of 9.6.1. Entities other than operators who discover undeclared or misdeclared dangerous goods should follow the reporting requirements of 9.6.2. These entities may include, but are not limited to, freight forwarders, customs authorities and security screening providers.





Dangerous Goods Regulations